

# Comprehensive Thesis Review & Actionable Recommendations

Technical Evaluation, Symbol Audits, and Quality Enhancements

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## Executive Summary

This report provides a comprehensive review of the Master's Thesis / Internship Report repository on **Isogeometric Analysis (IGA) and Reduced Order Modeling (ROM) for Parameterized Structural Dynamics**.

The overall manuscript is of **exceptional academic and technical quality**. The integration of NURBS-based IGA, Proper Orthogonal Decomposition (POD), Diffeomorphic Pull-back Mappings, and Energy-Conserving Sampling and Weighting (ECSW) hyper-reduction is mathematically rigorous, clearly derived, and validated by real-time client-side Digital Twin performance (**39× assembly speedup**, 20 FPS web visualizer).

This document outlines a high-level technical overview of the engineering pipeline, targeted chapter notes, mathematical symbol audits, and actionable recommendations for manuscript refinement.

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## 1 Technical Overview & Architectural Framework

### 1. Core Engineering Bottlenecks

Evaluating the transient dynamic response of structural components (such as a 2D notched cantilever beam) under varying geometric design parameters  $\boldsymbol{\mu} = (r, x_c)^T$  presents two major computational challenges:

1. **Online Remeshing & Assembly Overhead:** Standard Finite Element Method (FEM) solvers require spatial remeshing of the physical domain  $\Omega(\boldsymbol{\mu})$  whenever parameters change, incurring high latency ( $\approx 35.10$  ms per step just for matrix assembly).
2. **Vector Space Incompatibility:** As geometry changes, snapshot vectors collected across different parameter settings lie in different physical function spaces ( $\mathcal{V}(\Omega(\boldsymbol{\mu}_1)) \neq \mathcal{V}(\Omega(\boldsymbol{\mu}_2))$ ), invalidating standard subspace projection methods like Proper Orthogonal Decomposition (POD) in physical space.

### 2. The 4-Pillar Mathematical Solution

To resolve these bottlenecks, the thesis establishes a unified 4-pillar methodology:

1. **Isogeometric Analysis (IGA):** Uses Non-Uniform Rational B-Splines (NURBS)  $R_{i,j}^{p,q}(\boldsymbol{\xi})$  for both geometry representation and displacement field approximation  $(u, v)$ . Shape parameterization is achieved by updating control points  $\mathbf{P}_{i,j}(\boldsymbol{\mu})$  while keeping knot vectors and element connectivity fixed.

2. **Diffeomorphic Pullback Mapping ( $\hat{\Omega}$ ):** Maps the governing PDEs from the parameter-dependent physical domain  $\Omega(\boldsymbol{\mu})$  back to a static reference configuration  $\hat{\Omega}$  via a smooth mapping  $\boldsymbol{\Psi}(\boldsymbol{\xi}; \boldsymbol{\mu})$ . All integrals and gradients are pulled back using the coordinate Jacobian  $\mathbf{J}(\boldsymbol{\xi}; \boldsymbol{\mu})$  and its determinant  $J_{\boldsymbol{\mu}} = \det(\mathbf{J})$ . Because reference basis functions  $\hat{R}_a(\boldsymbol{\xi})$  are static, all snapshot vectors reside in the **same vector space**  $\mathcal{V}_0(\hat{\Omega})$ , enabling valid POD-SVD basis extraction.
3. **POD-Galerkin Subspace Reduction:** Offline Singular Value Decomposition (SVD) on reference displacement snapshots extracts an optimal  $r$ -dimensional reduced basis  $\boldsymbol{\Phi} \in \mathbb{R}^{N \times r}$  ( $r \ll N$ ), projecting the  $N$ -dimensional equations of motion down to reduced coordinates  $\mathbf{q}(t) \in \mathbb{R}^r$ :

$$\mathbf{M}_r \ddot{\mathbf{q}}(t) + \mathbf{C}_r \dot{\mathbf{q}}(t) + \mathbf{K}_r(\boldsymbol{\mu}) \mathbf{q}(t) = \mathbf{F}_r(t) \quad (1)$$

4. **Hyper-Reduction via ECSW:** Bypasses full-mesh integration of parameter-dependent stiffness matrices  $\mathbf{K}_r(\boldsymbol{\mu})$  by selecting a minimal active element subset  $\Omega_{\text{ECSW}} \subset \hat{\Omega}$  and positive weights  $w_e$  via Non-Negative Least Squares (NNLS). Operator assembly is reduced from 35.10 ms down to 0.90 ms (**39 $\times$  speedup**), preserving energy conservation and exact natural frequencies ( $\omega_{n,\text{ROM}} = 1.107$  rad/s vs.  $\omega_{n,\text{FOM}} = 1.108$  rad/s, deviation 0.09%).

### 3. Quantitative Performance Metric Comparison

Table 1: Benchmark Technical Metrics Comparison

| Metric / Feature        | Naive Physical Path                             | Mapped Reference + ECSW Path                            |
|-------------------------|-------------------------------------------------|---------------------------------------------------------|
| Mesh Topology           | Spatial remeshing on $\Omega(\boldsymbol{\mu})$ | Static reference mesh on $\hat{\Omega}$                 |
| Snapshot Space          | Incompatible across $\boldsymbol{\mu}$          | Unified space $\mathcal{V}_0(\hat{\Omega})$ (valid POD) |
| Operator Assembly Time  | 35.10 ms                                        | <b>0.90 ms (39<math>\times</math> speedup)</b>          |
| Total Step Latency      | > 40.00 ms                                      | <b>1.30 ms</b> (Interactive budget)                     |
| Graphics Refresh Rate   | $\approx 2 - 5$ FPS                             | <b>20 FPS</b> (Real-time Digital Twin)                  |
| Natural Frequency Error | Baseline                                        | <b>0.09%</b> (Exact spectrum saved)                     |

## 2 Chapter-by-Chapter Technical Notes & Review

### Chapter 1: Introduction & State of the Art

- **Strengths:** Section 1.2 (*Conceptual Parallels to Data Compression*) is a standout addition. Comparing CAD/IGA to vector graphics vs. raster grids, POD to spectral audio compression, and ECSW to optimal sensor placement builds strong intuition.
- **Recommendation:** In Section 1.4 (*Hyper-Reduction Methods*), explicitly contrast ECSW with the Discrete Empirical Interpolation Method (DEIM / MDEIM). Since DEIM papers are cited in `references.bib` ([?], [?]), articulating why ECSW was chosen for structural dynamics (preserving symmetry, system structure, and energy conservation) adds academic depth.

## Chapter 2: Theory: Problem Definition

- **Strengths:** Clean formulation of continuum mechanics, balance of linear momentum, Cauchy stress tensor, small strain tensor, and boundary conditions.
- **Symbol Inconsistency 1 (Notch Parameter):** In Figure 2.1 (TikZ schematic and caption), the notch depth/radius is labeled with the letter  $d$ . However, in equations, text, algorithms, and the Nomenclature, the design parameter vector is defined as  $\boldsymbol{\mu} = (r, x_c)^T$  using  $r$  for notch radius.
- **Symbol Inconsistency 2 (Reference Domain):** Section 2.1 defines the initial undeformed domain as  $\Omega_0$ . In Chapters 3, 4, 5, and the Nomenclature, this domain is denoted as  $\hat{\Omega}$ .
- **Action:** Change  $d \rightarrow r$  in Figure 2.1, and add a sentence in Section 2.1 linking  $\Omega_0$  to  $\hat{\Omega}$  ( $\hat{\Omega} \equiv \Omega_0$ ).

## Chapter 3: Discretization: Isogeometric Analysis (IGA)

- **Strengths:** Detailed derivation of Cox-de Boor recursion, B-splines, NURBS, spatial derivative evaluation, and B-matrix formulation. Section 3.2 gives a brilliant explanation of why spatial remeshing breaks vector space compatibility for SVD basis extraction.
- **Symbol Collision ( $\mathbf{d}$ ):** The symbol  $\mathbf{d}(\boldsymbol{\mu})$  is used for the parameter-dependent geometric mapping displacement ( $\mathbf{x} = \hat{\mathbf{x}} + \mathbf{d}(\boldsymbol{\mu})$ ), while  $\mathbf{d}_a$  in the Nomenclature denotes control node degrees of freedom.
- **Action:** Rename mapping field to  $\mathbf{u}_{\text{geom}}(\boldsymbol{\mu})$  or  $\mathbf{v}_{\text{map}}(\boldsymbol{\mu})$  to avoid confusion with control point displacement DOFs.

## Chapter 4: Simulation: Full Order Model (FOM)

- **Strengths:** Solid derivation of semi-discrete equations of motion and Generalized- $\alpha$  dynamic time integration.
- **Recommendation:** Explicitly state the high-frequency dissipation spectral radius  $\rho_\infty \in [0, 1]$  used to determine the algorithmic damping parameters  $(\alpha_m, \alpha_f, \beta, \gamma)$ . Standard choice is  $\rho_\infty = 0.8$  or  $1.0$ .

## Chapter 5: Reduced Order Modeling & Real-Time Digital Twin

- **Strengths:** Comprehensive coverage of SVD basis extraction, Galerkin projection, ECSW hyper-reduction, and client-side MVC Web application architecture.
- **Visual Ambiguity ( $\hat{\Omega}$  vs.  $\hat{\Omega}$ ):** The ECSW active element subset is written as  $\hat{\Omega}$  (wide hat), whereas the full reference domain is  $\hat{\Omega}$  (regular hat). In print, these look virtually identical.
- **Action:** Rename the active element subset to  $\Omega_{\text{ECSW}}$  or  $\hat{\Omega}_E$  in Chapter 5 and the Nomenclature. Add the singular value truncation energy threshold (e.g., 99.99%).

## Chapter 6 & Appendix: Conclusion & Benchmark Results

- **Strengths:** Concise summary of achievements against all research goals, backed by quantitative benchmark metrics (35.10 ms physical vs. 0.90 ms reference assembly).

- **Recommendation:** Suggest extending the pullback mapping framework to 3D solid mechanics or hyperelastic material models in future work.

### 3 Mathematical & Symbol Consistency Matrix

Table 2: Summary of Notation Audits and Standardizations

| Element                 | Current Thesis Notation     | Proposed Standard                                            | Target File / Location                                 |
|-------------------------|-----------------------------|--------------------------------------------------------------|--------------------------------------------------------|
| Reference Domain        | $\Omega_0$ & $\hat{\Omega}$ | $\hat{\Omega}$ (note $\Omega_0 \equiv \hat{\Omega}$ )        | thesis/chapters/<br>2_theory/index.tex                 |
| Notch Radius            | $d$ (in Fig 2.1) & $r$      | $r$ everywhere                                               | thesis/chapters/<br>2_theory/index.tex                 |
| ECSW Reduced Domain     | $\hat{\Omega}$              | $\Omega_{\text{ECSW}}$ or $\hat{\Omega}_E$                   | thesis/chapters/<br>5_rom/index.tex                    |
| Mapping Field           | $\mathbf{d}(\mu)$           | $\mathbf{u}_{\text{geom}}(\mu)$ or $\mathbf{u}_{\text{map}}$ | thesis/chapters/<br>3_discretization/<br>index.tex     |
| Parametric Parent Space | $\hat{\Omega} = [0, 1]^2$   | $\hat{\Omega}_{\text{param}} = [0, 1]^2$                     | thesis/chapters/<br>0_frontmatter/<br>nomenclature.tex |
| Natural Frequency       | $\omega_{n,\text{FOM}}$     | Add $\omega_n$ to Nomenclature                               | thesis/chapters/<br>0_frontmatter/<br>nomenclature.tex |

### 4 Actionable Priority Checklist

- ☐ **Fix TikZ Figure 2.1:** Update dimension label  $d \rightarrow r$  in `thesis/chapters/2_theory/index.tex`.
- ☐ **Unify Reference Domain:** Add clarifying sentence linking  $\Omega_0$  to  $\hat{\Omega}$  in `thesis/chapters/2_theory/index.tex`.
- ☐ **Disambiguate ECSW Mesh:** Replace  $\hat{\Omega} \rightarrow \Omega_{\text{ECSW}}$  in `thesis/chapters/5_rom/index.tex`.
- ☐ **Update Nomenclature:** Add  $\hat{\Omega}_{\text{param}}$ ,  $\Omega_{\text{ECSW}}$ , and  $\omega_n$  to `thesis/chapters/0_frontmatter/nomenclature.tex`.
- ☐ **Specify Generalized- $\alpha$  Damping:** State spectral radius  $\rho_\infty$  in `thesis/chapters/4_simulation/index.tex`.